Quantum Reference Algebra for Information Flow

The physical interface

Complete Collected Volume

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Introduction

A positive control cost does not become a Lorentzian metric simply by being renamed. These three papers connect operator representation, actual propagation and physical actuation. The connection is conditional on the carrier and interaction: a per-species cone is not automatically a common cone, and a representation image is not the whole apparatus that realizes it.

A develops propagation and readout, B realizes protection and corrective control, and C follows representations into local contacts, material sources and finite autonomous execution. The separate Mass–Energy, Null Cluster and No View papers extend these questions; they remain companion papers rather than additional chapters in this volume.

Reading map

QR-A *Lorentzian Read-out from Local Unitary Propagation: Clifford Relations, Causal Cones and Spectral Mass.* Local unitary propagation produces per-species Clifford readout and causal cones. Speed alignment requires divergent accumulated coupling in the mediator model, or decay of relative-speed perturbations near a supplied fixed point; connectivity alone is insufficient.

QR-B *Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost.* Persistent control is realized by physical couplings. Protection, calibration, active recovery and field energy are compared under explicit collection and reference permissions.

QR-C *Retained Quantum History, Physical Sources and Receiver Renewal in Q-RAIF.* One autonomous path performs finite repeated readings and receiver updates while retaining the body and history. Local linear-current control also implements neutral-sector stress flows; clock work and the limits of total-source matching are explicit.

The papers retain their complete proofs, appendices and references, and their own pagination. The contents list PDF page numbers; PDF bookmarks also provide section-level navigation.